

Smart Housing: Finding the Right Home with Data

A Data-Driven Housing Recommendation System



Prepared for:
Nicole Johnson



Prepared by:
Hadi Bordbar



Central



Lively



Affordable



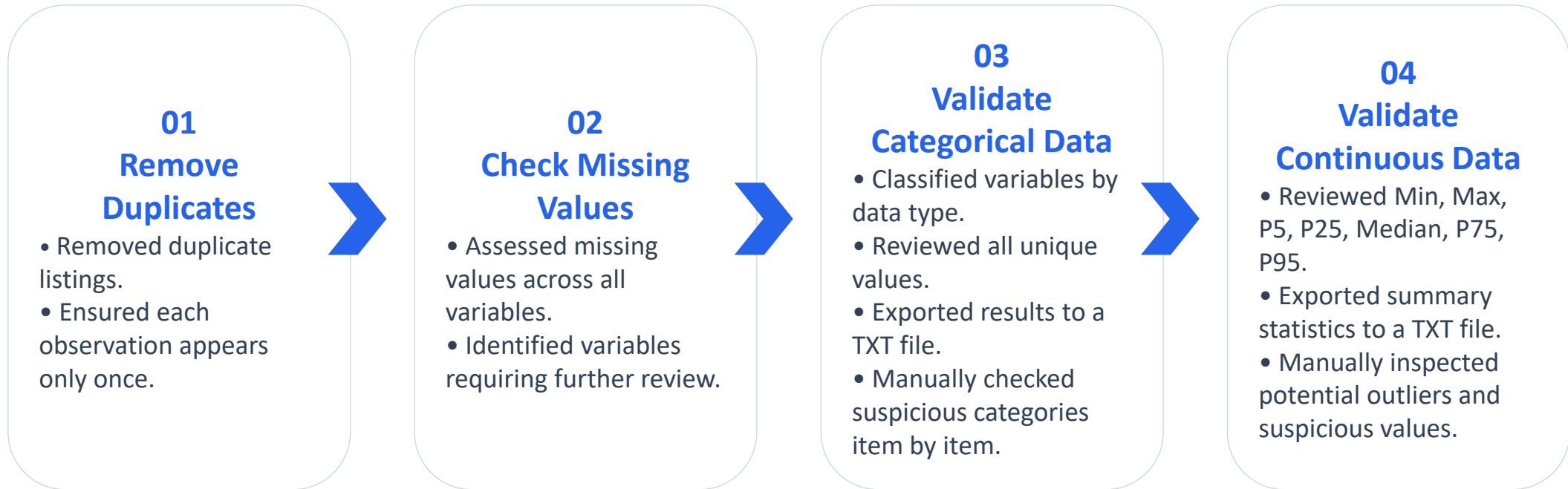
Available

>> neue fische

Contents

- Data Cleaning
- Assumptions & Methodology
- Defining “Lively & Central”
- Finding Suitable Districts
- Market Baseline for Selected Neighborhoods
- Room Price Comparison
- Houses

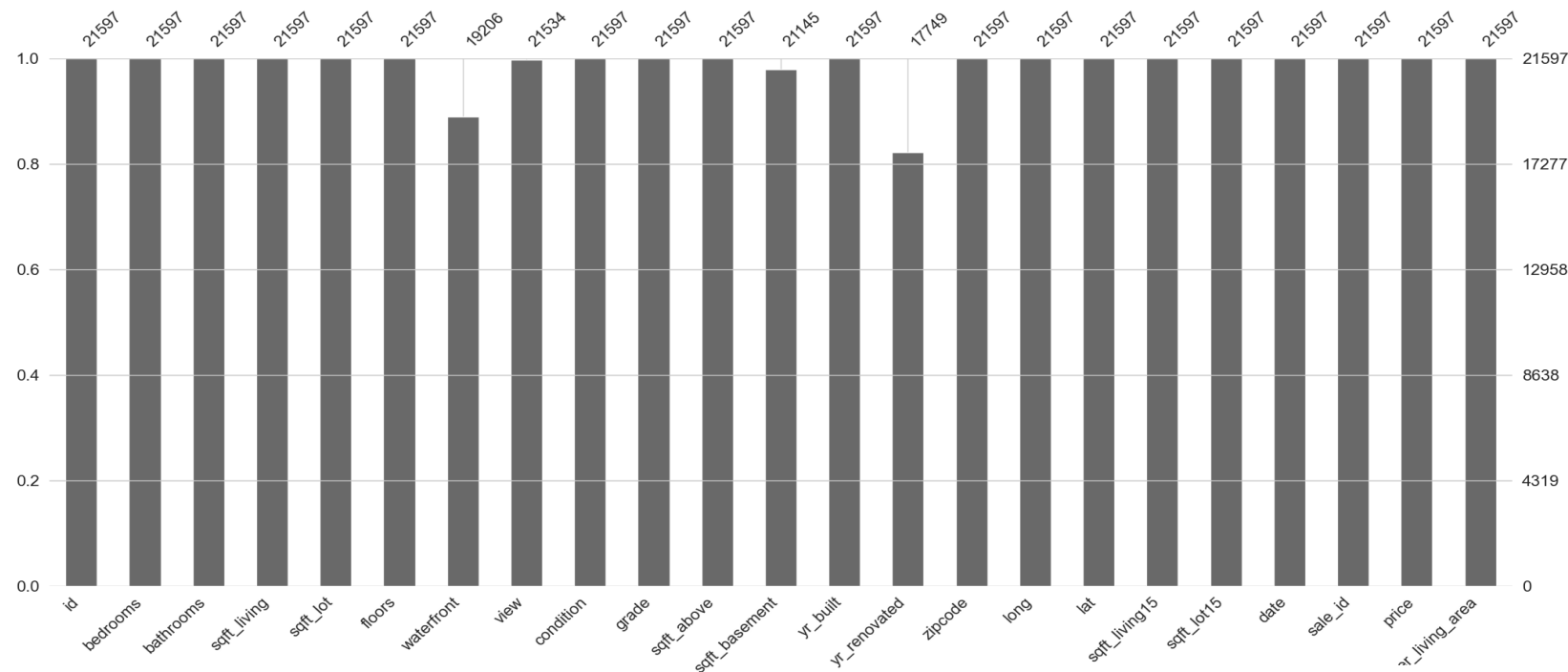
Data Cleaning & Quality Checks



Quality Assurance & Data Corrections

- ➔ Automated checks → Suspicious values identified → Manual inspection → Data correction / validation.
- ➔ Example correction: Renovation year values were found in an incorrect format
- ➔ Categorical Data: zipcode, bedrooms, bathrooms, floors, waterfront, view, condition, grade, yr_built, yr_renovated, date
- ➔ Continuous Data: sqft_living, sqft_lot, sqft_above, sqft_basement, long, lat, sqft_living15, sqft_lot15, price

Data Cleaning & Quality Assessment



Contin. variables

```
--- sqft_lot ---
0th percentile 520.0
1th percentile 1013.96
5th percentile 1800.8
25th percentile 5040.0
50th percentile 7618.0
75th percentile 10685.0
95th percentile 43307.20
99th percentile 213008.0
100th percentile 1651359.0

--- sqft_above ---
0th percentile 370.0
1th percentile 700.0
5th percentile 850.0
25th percentile 1190.0
50th percentile 1560.0
75th percentile 2210.0
95th percentile 3400.0
99th percentile 4370.0
100th percentile 9410.0
Missing values in sqft_basement 452
```

Categorical Variables

```
Unique values for floors:
    1.0, 2.0, 1.5, 3.0, 2.5, 3.5
Missing values in waterfront 2391

Unique values for waterfront:
    nan, 0.0, 1.0
Missing values in view 63

Unique values for view:
    0.0, nan, 3.0, 4.0, 2.0, 1.0
```

Assumptions & Methodology

Assumptions

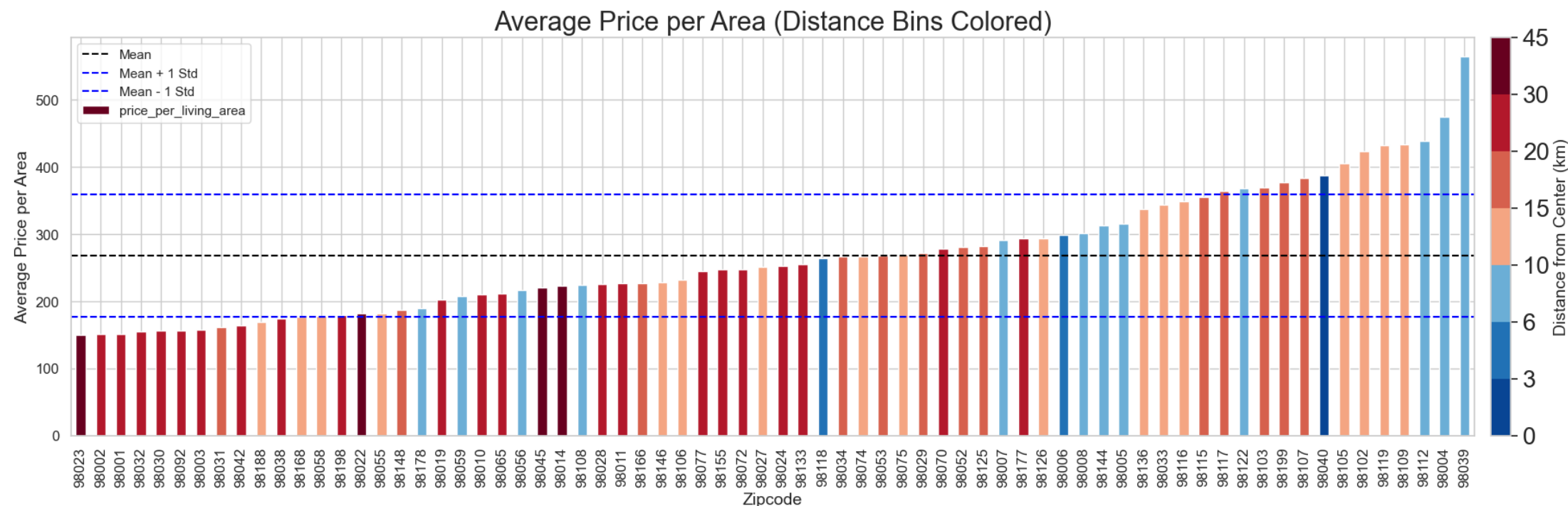
- Districts located less than 6 km from the city center offer a lively environment.
- The number of rooms represents the size of the flat.
- The fluctuation in the price of a flat per room depends on the neighborhood, the construction or renovation years, and the overall condition of the property.

Methodology

- Define “Lively & Central”
- Identify Suitable Districts
- Establish the Market Baseline
- Compare by Number of Rooms
- Benchmark Individual Listings
- Shortlist Suitable Flats

District Selection Based on Location & Price

Centrality is measured as the straight-line geographic distance (km) between each ZIP-code center and the defined city center.

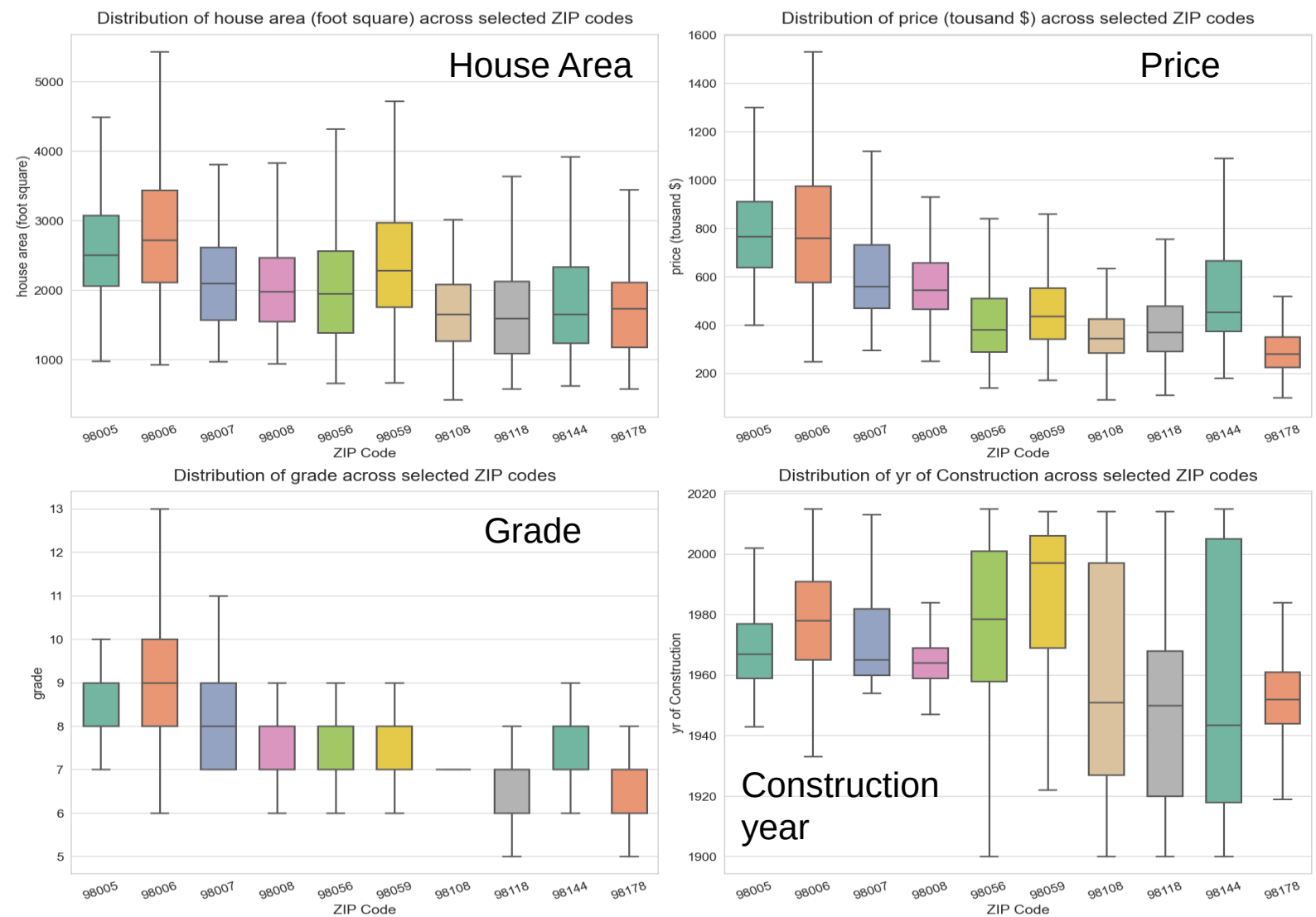


Districts ranked by average price per sq. ft. of living area; bar color indicates distance from the center.

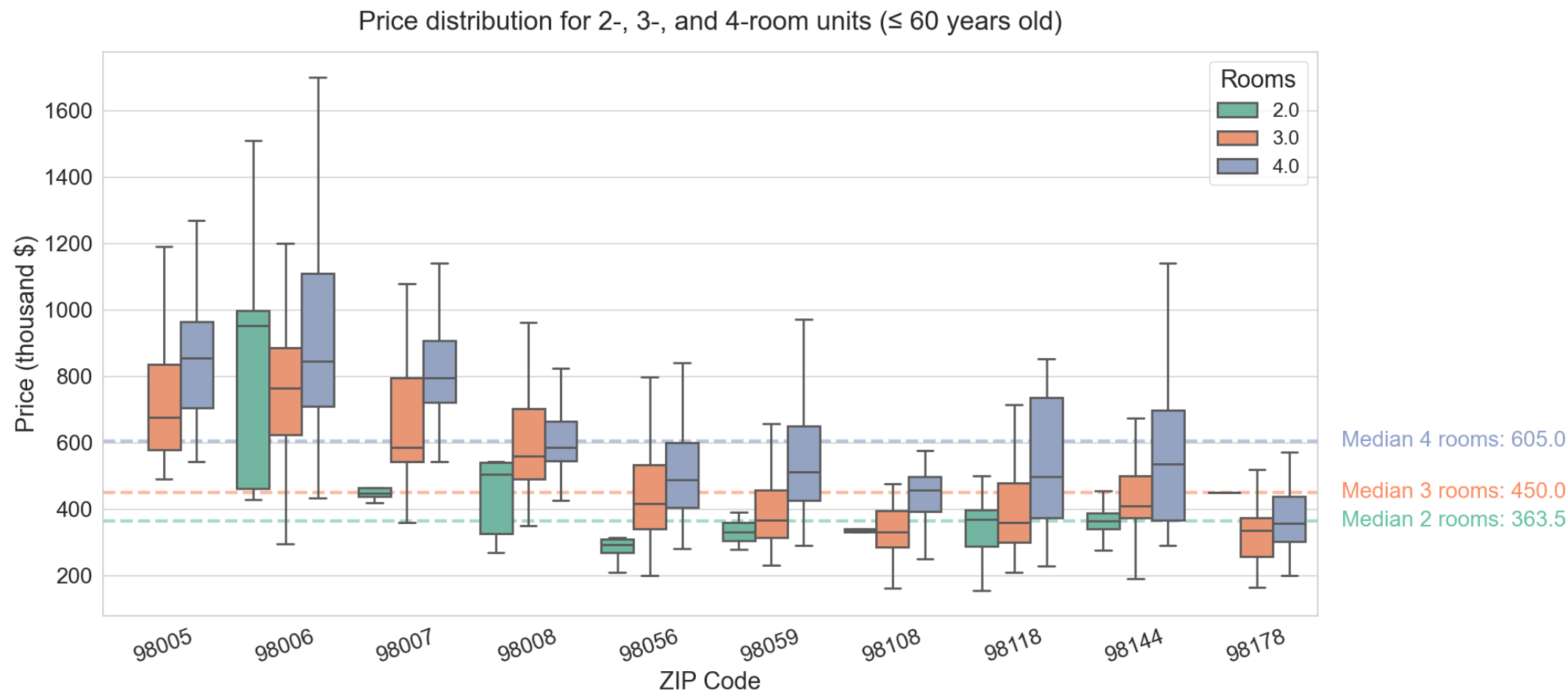
ZIP Codes Selected for Further Analysis

98178 | 98059 | 98056 | 98108 | 98118 | 98007 | 98006 | 98008 | 98144 | 98005

Distribution of Key Housing Characteristics in Selected Districts

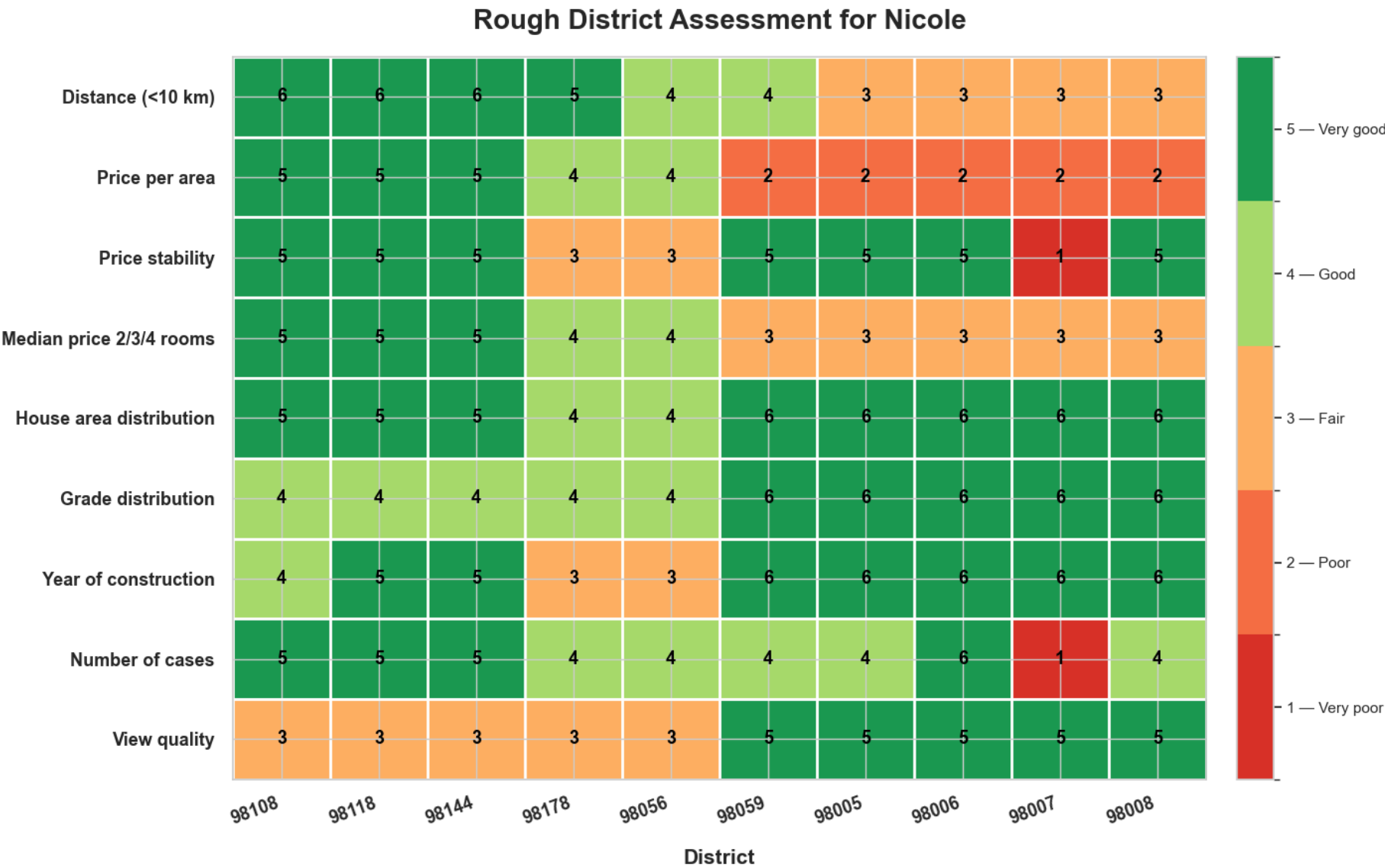


House Price Comparison by Number of Rooms



Price distribution by room count and district; horizontal lines represent the overall market median for each room category.

Selected Neighborhoods: Multi-Criteria Comparison



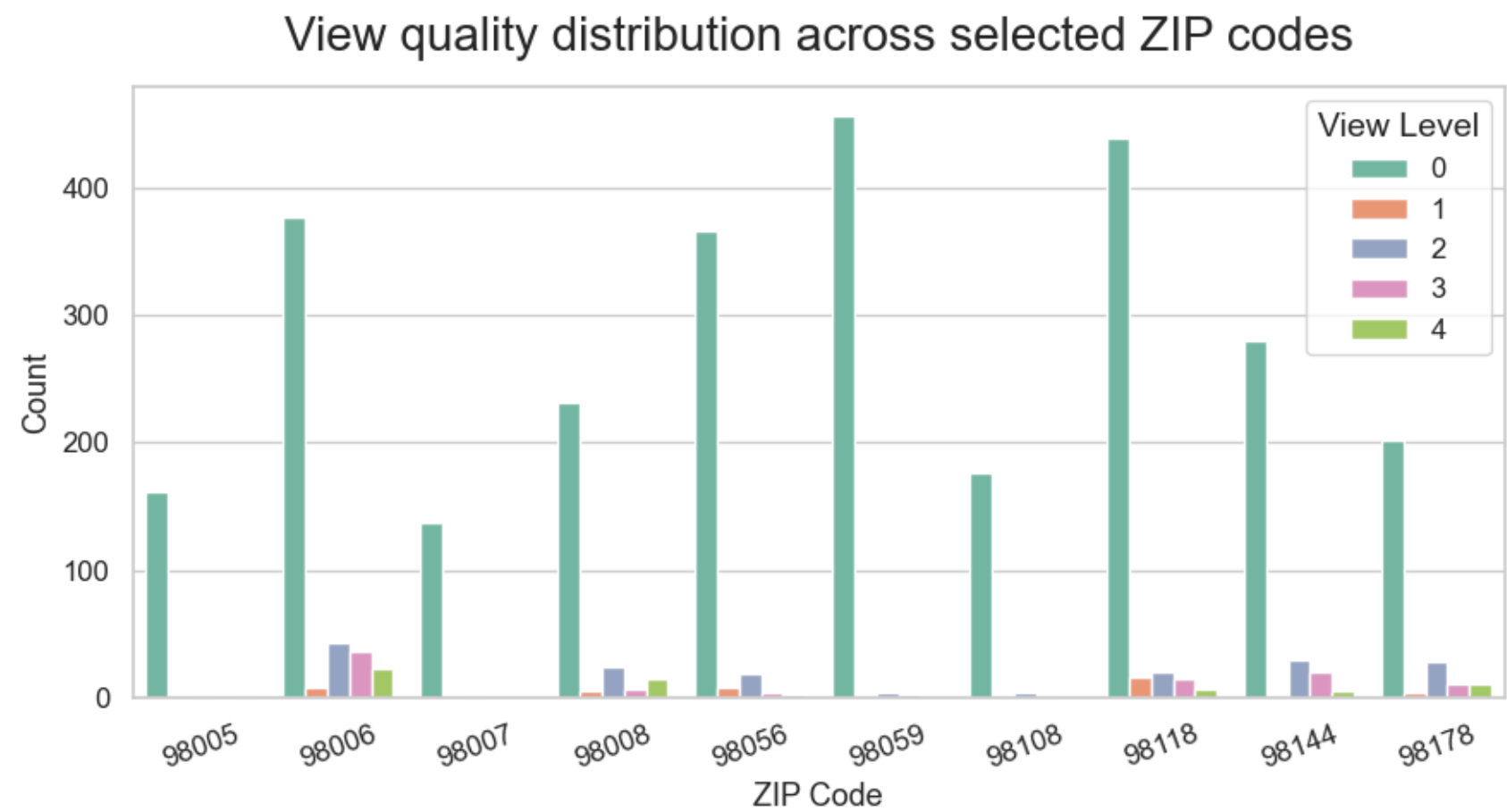
Recommended Properties Based on Our Analysis

Flats/House with Two Bedrooms											
id	zipcode	bedrooms	sqft_living	price	view	grade	yr_built	yr_renovated	waterfront	date	
7548301056	98144	2.000	1340.000	345000.000	0.000	8	2008	0.000	0.000	2014-06-09	
1498300775	98144	2.000	930.000	355000.000	0.000	8	2007	0.000	0.000	2014-06-24	
7548301044	98144	2.000	1320.000	342500.000	0.000	8	2008	0.000	0.000	2014-07-10	
1604601572	98118	2.000	860.000	345000.000	0.000	9	2009	0.000	0.000	2014-09-05	
567000392	98144	2.000	920.000	363000.000	0.000	8	2009	0.000	0.000	2014-11-03	
6056111370	98108	2.000	1270.000	340000.000	0.000	8	2012	0.000	0.000	2014-11-24	
424049283	98144	2.000	830.000	345000.000	0.000	8	2009	0.000	0.000	2015-04-15	

Flats/House with Three Bedrooms											
id	zipcode	bedrooms	sqft_living	price	view	grade	yr_built	yr_renovated	waterfront	date	
6909200335	98144	3.000	1430.000	457500.000	0.000	8	1996	0.000	0.000	2014-06-24	
1604602050	98118	3.000	1610.000	460000.000	2.000	9	2005	0.000	NaN	2014-07-11	
1498300875	98144	3.000	1550.000	445000.000	0.000	8	2006	0.000	0.000	2014-08-14	
567000268	98144	3.000	1639.000	450000.000	0.000	8	2003	NaN	0.000	2014-08-21	
9320350130	98108	3.000	2330.000	453000.000	0.000	9	2004	0.000	NaN	2014-08-23	
5700000446	98144	3.000	1590.000	465000.000	0.000	8	2014	0.000	0.000	2014-10-29	
2394600157	98144	3.000	1530.000	460000.000	0.000	8	2008	0.000	0.000	2015-04-07	
8944550050	98118	3.000	2080.000	448500.000	0.000	8	2010	NaN	0.000	2015-05-08	

*Thank You
For Your Attention*

Selected Neighborhoods: View Quality



Selected Neighborhoods: Available Houses

